Johnny Zhang

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Education

University of California, Los Angeles Graduation June 2028

*B.S. **Electrical Engineering**, Computer Science Technical Breadth* *GPA: 3.95/4.0*

ECE Department Fast Track Honors Program (Top 20 students)

Skills

Hardware: Verilog, SystemVerilog, RISC-V, Vitis HLS, FPGA programming, AMD Vivado, computer architecture
Software: C++, C, Linux, Git, Python, LLVM, embedded systems, machine learning, data structures and algorithms
RF & Wireless: Spectrum sensing, software-defined radio (SDR), MATLAB, digital signal processing

Publications & Awards

- **Zhang, Johnny** et al. "RLESS: A Lightweight and Low-Latency Run Length Encoding-Based Spectrum Sensing Algorithm Implemented on RFSoC." Accepted to IEEE Military Communications Conference, 2026.
- C. Steigerwald, T. Zhao, B. Domae, **J. Zhang** and D. Cabric. "HI-WHIRLS: A Real-Time FPGA Architecture for Signal Labeling in Spectrum Sensing." Accepted to IEEE Aerospace Conference, 2026.
- T. Zhao, C. Steigerwald, **J. Zhang** and D. Cabric. "SCD: Segment-CNN-Decision for Signal Detection in Wideband Spectrum with Unknown Interference." IEEE Balkan Conference on Communications and Networking, 2026.
- **1st Place Winner**, NSF SpectrumX Student Data and Algorithms Competition
- **USA Computing Olympiad**: Gold Division, **USA Physics Olympiad**: 3x Semifinalist, Honorable Mention (2023)

Experience

Cognitive Reconfigurable Embedded Systems Lab Oct 2025 - Present

***Undergraduate Researcher** (Full Time Jun 2026 - Sep 2026)* *Los Angeles, CA*

- Designed a machine learning-based algorithm to detect transmissions in the 7 GHz band from raw I/Q samples using convolutional neural networks, achieving 89% accuracy in extreme low-SNR and unknown interference scenarios
- Built a signal classification pipeline on a Xilinx RFSoC ZCU216 and NVIDIA Jetson embedded platform, using a YOLO-based approach to continuously label BPSK, QPSK and QAM modulations from 1.6 Gbps of over-the-air data
- Retrained the YOLO model for INT8 quantization, implemented Fast Fourier Transform and pre-processing on the FPGA side to reduce runtime by 94% and achieve a real-time performance of 381 inferences per second

UCLA Summer Undergraduate Research Program Jun 2025 - Sep 2025

Research Intern *Los Angeles, CA*

- Developed a high-speed spectrum sensing algorithm for the 2.4 GHz band in C++, optimized with multithreading and efficient cache access patterns resulting in an 18x speedup over similar machine learning and DSP-based methods
- Programmed the algorithm onto a Xilinx RFSoC 4x2 FPGA using Verilog and high-level synthesis, achieving real-time performance with sub-millisecond latency while using less than 4% of the board's LUTs, block rams and DSP slices
- Used SystemVerilog to testbench individual modules, verified the overall system using integrated logic analyzer, signal generator and spectrum analyzer

Scale AI Feb 2025 - Aug 2025

Generative AI Intern *San Francisco, CA*

- Built high quality datasets for LLM reasoning on Codeforces Grandmaster and USACO Platinum-level competitive programming problems, significantly strengthening performance on algorithmic reasoning benchmarks
- Scraped data from GitHub to fine-tune GPT-5-based agentic models to autonomously generate Docker images and configure runtime environments, streamlining deployment pipelines and reducing manual setup time by up to 15%
- Designed and validated datasets of graduate-level physics problems to train large language models for Humanity's Last Exam, improving scores by 2.15x over the course of 5 months

Projects

RISC-V DSP extension core | *SystemVerilog, C, LLVM* Jun 2026 - Present

- Developed an extension core for 32-bit RISC processors to accelerate DSP applications, supporting saturating fixed-point arithmetic, complex numbers, SIMD and vector operations, implemented on a Zynq 7020 FPGA
- Built an LLVM backend to convert C programs into machine code for the CPU and extension core
- Developed testbenches using SystemVerilog, benchmarked performance of common DSP functions like FIR filters, FFT and coordinate rotation against a base RV32I processor, reduced clock cycle usage by up to 39%

Custom CPU | *Verilog, Assembly, C++* Mar 2024 - Present

- Designed a 16-bit CPU featuring a custom instruction set architecture, five-stage pipeline, branch predictor and hardware support for operating systems, implemented on a Xilinx Zynq 7020 FPGA
- Built an assembler using C++ to translate human-readable code into binary executables for the CPU